BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

CHEMISTRY PART-I — (Re-appear) — (VERSION A) — SECTION-A

Time Allowed: 20 minutes | **Maximum Marks:** 18 | **Paper Code:** BN-XI-XXII-II

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number in provided bubble sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. The number of moles in 22 g of CO_2 is/are?

(A) 0.5
(B) 1
(C) 1.5
(D) 2

Q2. Efficiency of a chemical reaction can be checked by calculating:

(A) Actual yield
(B) Theoretical yield
(C) Percentage yield
(D) None of these

Q3. The energy of one quantum is equal to:

(A) $h\nu$
(B) $1.5\ h\nu$
(C) $2\ h\nu$
(D) $2.5\ h\nu$

Q4. The maximum number of electrons in p-orbital are:

(A) 2
(B) 4
(C) 6
(D) 8

Q5. According to electronic configuration, the electron of H atom is accommodated in:

(A) 1s
(B) 2s
(C) 2p
(D) 3s

Q6. The shape of $SnCl_2$ is:

(A) Linear
(B) Trigonal
(C) Tetrahedral
(D) Angular

Q7. p-character in sp hybrid orbital is:

(A) 25%
(B) 33%
(C) 50%
(D) 75%

Q8. Oxygen molecule is paramagnetic due to the presence of:

(A) Paired electrons in σ bonding molecular orbitals
(B) Vacant σ* bonding molecular orbitals
(C) Unpaired electrons in π* 2p_y and π* 2p_z anti bonding molecular orbitals
(D) Paired electrons in π 2p_y and π 2p_z bonding molecular orbitals

Q9. The 1 atm pressure is equal to:

(A) 760 mm of Hg
(B) 760 torr
(C) 101325 Pa
(D) All of these

Q10. The spontaneous mixing of various non-reacting gases is due to:

(A) Diffusion
(B) Effusion
(C) Low density
(D) Gravity

Q11. The number of center of symmetry for a crystal is/are:

(A) 1
(B) 2
(C) 3
(D) 4

Q12. Dipole-dipole force is found in _____ molecules:

(A) Polar
(B) Non-polar
(C) Ionic
(D) None of these

Q13. According to Le Chatelier's Principle, pressure change is applicable for those gaseous reversible reactions in which volume of:

(A) Reactant < Product
(B) Reactant > Product
(C) Reactant = Product
(D) All of these

Q14. The oxidation state of Cr in $K_2Cr_2O_7$ is:

(A) +2
(B) +4
(C) +6
(D) +8

Q15. The standard electrode potential of SHE is arbitrary chosen as:

(A) 0 V
(B) 1 V
(C) 2 V
(D) 2 V

Q16. K_c and K_p are:

(A) Directly related
(B) Inversely related
(C) equal
(D) None of these

Q17. The molarity of 100 g of NaOH in 1 dm^3 of a solution is/are:

(A) 1
(B) 1.5
(C) 2
(D) 2.5

Q18. The example of completely miscible liquids is:

(A) Water and Benzene
(B) Water and Phenol
(C) Water and Alcohol
(D) Water and Ether

CHEMISTRY PART-I — (Re-appear) — SECTION-B

Time Allowed: 02:40 Hrs | **Maximum Marks:** 40

> Note: Note: Attempt section B & C accordingly.

Q2. (Marks 40) Q.No.2: Attempt any 10 parts of the following. All parts carry equal marks.

i. Why actual yield is less than theoretical yield?

ii. What are limiting and excess reagents?

iii. Differentiate between isomorphism and polymorphism.

iv. Calculate the mass of an electron in kg.

v. Can you give the reason that 4s orbital are written first than the 3d orbital.

vi. Write important points of VSEPR theory.

vii. Sigma bond is stronger than pi bond, why?

viii. Calculate the density in g dm^{-3} of SO_2 at 0°C and 1 atm pressure.

ix. Explain difference between in B.P of water at sea level and at high mountains.

x. What is the relationship between K_c and K_p.

xi. What are liquid crystals?

xii. Discuss dry cell.

xiii. What is common ions effect?

SECTION-C

Maximum Marks: 27

Note: Note: Attempt any (THREE) questions of the following. All questions carry equal marks.

Q03. Q.NO.03. a) What is law of mass action, explain it in detail.

b) Calculate Kc for the synthesis of ammonia at 500°C, Kp= 1.44×10^{-5} according to the balance chemical equation: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

Q04. Explain Elevation of B.P in detail. (Boiling point)

Q05. Q.NO.05. a) What is Born Haber cycle?

b) Applying Hess's law calculate ΔH^0 for the sublimation of one mole of iodine from the following equation: i. $H_2(g) + I_2(s) \rightarrow 2HI\ (g)$ ΔH^0=51.8 kJ/mol ii. $H_2(g) + I_2(g) \rightarrow 2HI\ (g)$ ΔH^0=-10.5 kJ/mol

Q06. Q.NO.06. a) Calculate the radius of Hydrogen atom.

b) How much energy is released when an electron in hydrogen atom jump from n_2=3 to n_1=1?

GROUP – II

CHEMISTRY (PART-I) — (Fresh) — (Version-A) — SECTION –A

Time Allowed: 20 Min | **Maximum Marks:** 18 | **Paper Code:** BN-XI-XXII-II

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Efficiency of Chemical reactions can be checked by calculating.

(A) Actual yield
(B) Theoretical yield
(C) Percentage yield
(D) Amount of reactant unused

Q2. For reaction (a)H_2 + (a)Cl_2 ⟶ (a)2HCl the correct equivalency is

(A) 1mole of $H_2 \cong$ 1 mole of HCl
(B) 22.4 dm^3 of $H_2 \cong$ 44.8 dm^3 of Cl_2
(C) 22.4 dm^3 $Cl_2 \cong$ 44.8 dm^3 of HCl
(D) 1mole of $Cl_2 \cong$ 22.4 dm^3 of HCl

Q3. Which of the following sub shell do not exist?

(A) 1P
(B) 5d
(C) 6f
(D) 1S

Q4. When electrons undergo transition from N→L shells, then x-rays produced are

(A) Lα
(B) Lβ
(C) Lγ
(D) Lδ

Q5. If 2.65 g of Na_2CO_3 is dissolved in 250ml of H_2O, the molarity of solution will be:

(A) 0.1
(B) 1.0
(C) 2.0
(D) 0.2

Q6. Solids having only Van der Waals forces between their particles are _____ solids.

(A) Network
(B) Metallic
(C) Ionic
(D) Molecular

Q7. The volume of a gas is inversely proportional to pressure provided that

(A) Temperature is constant
(B) Pressure is constant
(C) Number of moles of gas is constant
(D) Both temperature and number of moles of gas are constant

Q8. In ideal gas equation (PV = nRT) V represents

(A) Total volume of gas molecules and free space
(B) Effective molar volume of gas molecules
(C) Volume of free space
(D) Volume of gas molecules

Q9. Rate of evaporation of a liquid depends on

(A) Temperature
(B) Intermolecular forces
(C) Surface area
(D) All of these

Q10. The highest ionic character is found in:

(A) HI
(B) HBr
(C) HCL
(D) HF

Q11. Kc for a reversible reaction is equal to

(A) Kr/Kf
(B) Kf/Kr
(C) Kr × Kf
(D) Kf/2Kr

Q12. Gel is an example of:

(A) True solution
(B) Colloidal solution
(C) Suspension
(D) Solid

Q13. Rate law for a chemical reaction is $dx/dt = K[A]^1 [B]^{1.5}$ the order of reaction will be

(A) 1 (B) 2

(C) 2.5 (D) 1.5

Q14. The unit of rate constant (K) for first order reaction is

(A) S^{-1} (B) $moldm^{-3}s^{-1}$

(C) $dm^3mol^{-1}s^{-1}$ (D) $(dm^3)^2mol^{-2}s^{-1}$

Q15. In face centered cubic lattice each point at corner is shared by unit cells:

(A) 6 (B) 8

(C) 2 (D) 4

Q16. When a liquid is dispersed in another liquid then colloid is called

(A) Fog (B) Aerosol

(C) Emulsion (D) Gel

Q17. The enthalpy changes which is always negative

(A) Enthalpy of solution (B) Enthalpy of formation

(C) Enthalpy of reaction (D) Enthalpy of neutralization

Q18. The oxidation state of Br in $HBrO_2$ is

(A) +1 (B) +3

(C) +5 (D) -3

CHEMISTRY (PART-I) — (Fresh) — SECTION –B

Time Allowed: 2:40Hrs | **Maximum Marks:** 40

Note: Note: Attempt section B & C accordingly.

Q2. (Marks 40) Q.No 2. Attempt any TEN parts of the following. All parts carry equal marks.

i. What is the difference between isomorphism and polymorphism? Give examples.

ii. Define and explain Pauli's exclusion principle with one example.

iii. Differentiate between sigma and pi bonds.

iv. Write any four properties of plasma state.

v. Explain the role of hydrogen bonding in structure of ice.

vi. Calculate solubility of $PbSO_4$ at 25°C if its Ksp is 1.6×10^{-8}

vii. Explain why the sum of PK_a and PK_b is always equal to 14.

viii. Calculate pH and pOH of 0.005 mol dm^{-3} solution of H_2SO_4

ix. Write main assumptions of collision theory of reaction rates.

x. Define solubility. Explain the effect of pressure on the solubility of a substance.

xi. What is pressure volume work? How it is related to first law of thermodynamics?

xii. Write any four methods for the prevention of corrosion.

xiii. Define a cubic unit cell. Also write the three types of cubic unti cells with proper diagrams.

SECTION –C

Maximum Marks: 27

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) In an experiment 01g of H_2 was reacted with excess of O_2. Calculate the actual yield (in grams) of H_2O if percentage yield was 60%.

b) What is the PH of solution, if $[H_3O]^+$ is 6.7 x 10^{-4} M?

Q4.

a) Draw and explain molecular orbital diagram of O_2

b) Define acids and bases according to Lewis's concept. Give two examples of each.

Q5.

a) On the basis of Bohr's atomic model, derive the radius of hydrogen atom.

b) What is Born Haber cycle. Explain it with one example in detail.

Q6.

a) Define catalysis. Write it's types with definitions and one example of each type.

b) Balance the following redox equation by oxidation number method. $Fe + V_2O_3 \longrightarrow Fe_2O_3 + VO$